

Submission Summary

Summary of Completed and Incomplete Work & Prioritization Decisions

1. Summary of Completed Work

The following components have been fully implemented to a professional, production-ready standard:

Component / Deliverable	Implementation Standard & Highlights
Data Ingestion & Cleaning	Standardized currency strings, handled missing values via robust median imputation, removed extreme outliers (>€10k/night), and validated coordinate boundaries within Barcelona. Output persisted in Parquet.
DuckDB Data Warehouse	Built a Star Schema (1 Fact, 3 Dimensions) optimized for OLAP aggregations. Implemented dynamic views for host portfolios and listing economics (San Francisco occupancy model).
Statistical Analysis	Conducted non-parametric hypothesis testing (Mann-Whitney U, Kruskal-Wallis H) across 5 core hypotheses (H1 to H5), verifying assumptions and computing effect sizes (Cohen's d, CLES, η^2).
Predictive Pricing Model	Trained an XGBoost Regressor achieving an R² of 0.857 and MAE of €50.36 . Integrated SHAP TreeExplainer for full model explainability. Saved serialized model artifacts in the <code>/models</code> directory.
Content-Based Recommender	Engineered a Cosine-Similarity based recommendation system using one-hot encoded amenity vectors, solving the cold-start problem for never-booked listings.
Generative AI Integration	Connected Groq's Llama-3.3-70b-versatile model to interpret XGBoost predictions and positive/negative SHAP drivers, returning personalized, written strategic advice to hosts.
Interactive Streamlit Dashboard	Created a responsive 5-tab dashboard displaying Market Overview, Geographic Analysis, Host Concentration, the AI Price Advisor tool, and AI-generated briefings.
Automated Unit Tests	Implemented <code>pytest</code> suite verifying data cleaning functions, coordinates, and database schema integrity.
Docker Containerization	Provided a <code>Dockerfile</code> and <code>docker-compose.yml</code> to package and run the application with single-command reproducibility.

2. Summary of Incomplete Work & Prioritization Decisions

In accordance with Section 11 of the assignment, the following items were intentionally left incomplete or scoped out. This is a conscious engineering trade-off to prioritize quality and depth over superficial completion.

2.1 Cross-City Comparisons (Section 12 of Final Report)

- **What was left out:** Ingesting and modeling multiple cities (e.g., London or Paris).
- **Prioritization Rationale:** The evaluation rubric places heavy weight on problem-solving depth, statistical rigor, and clear communication. Spreading development effort across multiple cities within a one-week timeline would have resulted in a generic, surface-level analysis. Instead, we chose to execute a deep-dive on the Barcelona market, delivering comprehensive statistical testing, advanced ML explainability, a custom recommender system, and GenAI pricing advisors.
- **Mitigation:** The code is structured architecturally to accept new city datasets with zero changes via environment variables.

2.2 dbt Model Integration

- **What was left out:** Creating a standalone dbt project to manage warehouse transformations.
- **Prioritization Rationale:** For a single-city analytical dataset of this size (~15,000 listings), introducing dbt adds significant deployment, orchestration, and compilation overhead with no runtime performance benefit.
- **Mitigation:** Database migrations, table definitions, and star-schema views are natively managed in Python using raw SQL strings via DuckDB connections.

2.3 Continuous Integration (CI/CD) and Automated Re-runs

- **What was left out:** Deployed Apache Airflow DAGs and automated drift monitoring (e.g., Evidently AI).
- **Prioritization Rationale:** These are production-ready MLOps components that require active cloud environments to run. Building mock infrastructure locally would take development time away from visual storytelling and analytical depth.
- **Mitigation:** A complete MLOps roadmap (including Airflow design sketches and Evidently AI drift-detection triggers) is detailed in Section 14 of the main report.

3. Assumptions & Decisions Log Summary

All engineering decisions (imputations, log-transformations, DuckDB schema choice, and Responsible AI mitigation for Nou Barris pricing bias) are fully documented in:

- reports/assumptions_and_decisions_log.md

4. AI Usage Disclosure

A full log of AI tools used, prompts applied, output validation methods, and suggestions that were explicitly rejected is documented in:

- Appendix A of the Final Report